

Multivariable Calculus Practice Questions

Instructions

Answer the following 28 questions covering all key topics from multivariable calculus. Show all steps for full understanding. Use appropriate formulas and theorems.

1. Find the parametric equations of the line through the point $(1, 2, 3)$ and parallel to the vector $\langle 4, -1, 2 \rangle$.
2. Determine the equation of the plane passing through the point $(0, 1, -2)$ with normal vector $\langle 2, -3, 1 \rangle$.
3. Find the angle between the planes $x + 2y - z = 4$ and $2x - y + 3z = 1$.
4. Find the length of the curve $\vec{r}(t) = \langle t, t^2, t^3 \rangle$ from $t = 0$ to $t = 1$.
5. Find the tangent vector to the curve $\vec{r}(t) = \langle \cos t, \sin t, t \rangle$ at $t = \pi$.
6. Convert the Cartesian equation $x^2 + y^2 = 4$ into polar form.
7. Compute the area inside the cardioid $r = 1 + \cos \theta$.
8. Find the arc length of the polar curve $r(\theta) = 2\theta$ for $0 \leq \theta \leq \pi$.
9. Find the gradient of $f(x, y, z) = x^2y + yz^3$ at the point $(1, 1, 1)$.
10. Compute the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\langle 3, 4 \rangle$.
11. Use the chain rule to compute $\frac{dz}{dt}$ if $z = xy$, where $x = \sin t$, $y = e^t$.
12. Evaluate the double integral $\int_0^2 \int_0^x (x^2 + y^2) dy dx$.
13. Compute the triple integral $\iiint_E z dV$, where E is the box $0 \leq x \leq 1$, $0 \leq y \leq 2$, $0 \leq z \leq 3$.
14. Convert the integral $\int_0^{2\pi} \int_0^1 r^2 dr d\theta$ to Cartesian coordinates and evaluate.
15. Let $x = u + v$, $y = u - v$. Find the Jacobian $\frac{\partial(x,y)}{\partial(u,v)}$.
16. Use a suitable change of variables to evaluate $\int_0^2 \int_0^{2y} (x - y)^2 dx dy$.
17. Let $\vec{F}(x, y) = \langle y, -x \rangle$. Compute the line integral of $\vec{F}$ along the unit circle counterclockwise.
18. Determine whether the vector field $\vec{F}(x, y) = \langle 2xy, x^2 \rangle$ is conservative.
19. Find a potential function for the conservative vector field $\vec{F}(x, y) = \langle 3x^2 + 2y, 2x + 4y^3 \rangle$.

20. Compute the divergence of $\vec{F}(x, y, z) = \langle x, y, z \rangle$.
21. Compute the curl of $\vec{F}(x, y, z) = \langle yz, xz, xy \rangle$.
22. Find the surface area of the part of the plane $z = x + y$ that lies over the region $0 \leq x \leq 1$, $0 \leq y \leq 1$.
23. Compute the flux of $\vec{F}(x, y, z) = \langle 0, 0, z \rangle$ through the surface $z = 4 - x^2 - y^2$, over the disk $x^2 + y^2 \leq 4$.
24. Use Green's Theorem to evaluate $\oint_C (x^2 - y) dx + (xy + x) dy$ where C is the boundary of the region bounded by $x = 0$, $x = 1$, $y = 0$, $y = 1$.
25. Use Stokes' Theorem to evaluate the line integral $\oint_C \vec{F} \cdot d\vec{r}$, where $\vec{F}(x, y, z) = \langle y, -x, z \rangle$ and C is the boundary of the upper hemisphere of $x^2 + y^2 + z^2 = 1$.
26. Use the Divergence Theorem to compute the flux of $\vec{F}(x, y, z) = \langle x, y, z \rangle$ out of the surface of the unit cube $[0, 1]^3$.
27. Find and classify the critical points of $f(x, y) = x^3 - 3xy^2$.
28. Use Lagrange multipliers to find the max and min of $f(x, y) = xy$ subject to the constraint $x^2 + y^2 = 1$.